

Peer Comparison Methodology Reference

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Audience: Diversion Support Team, Clinical Leadership

Overview

Peer comparison analysis is a statistical method used to identify medication dispensing and administration patterns that deviate significantly from a defined peer exemplar cohort. This reference document defines the methodology, cohort selection criteria, metrics, and interpretation guidelines used by the Diversion Support Team.

Peer Cohort Definition

Peer Exemplar Cohort: A group of 15-30 licensed healthcare providers who share key characteristics with the subject employee, used as a statistical baseline for comparison.

Cohort Selection Criteria

Criterion	Definition	Rationale
Licensed Role	RN, LPN, Tech (same as subject)	Role determines scope of practice and medication access
Unit/Department	Same as subject (e.g., Med-Surg, ICU, ED)	Patient acuity and medication utilization varies by unit
Shift	Same shift as subject (Day/Evening/Night)	Medication utilization patterns differ by shift
Time Period	Minimum 60 days of concurrent activity	Sufficient data for meaningful statistical comparison
Data Quality	Providers with complete record documentation	Excludes providers with intermittent charting or gaps

Comparison Metrics

1. Waste Rate and CII Waste Rate

Definition: Percentage of medications dispensed that were wasted (not administered to patient).

Formula:

Waste Rate = (Total Waste Transactions / Total Dispense Transactions) * 100%
CII Waste Rate = (CII Waste Transactions / Total CII Dispense Transactions) * 100%

Interpretation: Rates significantly above peer group (>2 standard deviations) may indicate inappropriate dispensing, poor technique, or concerning patterns.

2. Null Transaction Rate

Definition: Percentage of dispensed medications with no corresponding administration record.

Formula:

Null Rate = (Dispense with No Admin Record / Total Dispense) * 100%

Interpretation: Elevated null rates suggest medications dispensed but not given, potentially diverted.

3. Pain-Score Correlation Index

Definition: Percentage of opioid administrations preceded by documented pain assessment at threshold level (pain >= 4/10).

Calculation:

Correlation Index = (Opioid Admins with Pain >= 4 / Total Opioid Admins) * 100%

Peer Baseline: Typical range 85-95% for nurses in med-surg and critical care settings.

Interpretation: Scores significantly below peer baseline (<70%) suggest opioids administered without clear clinical justification.

4. Off-Shift Activity Percentage

Definition: Percentage of medication transactions occurring outside scheduled shift hours.

Interpretation: Elevated off-shift activity may indicate unauthorized access or system manipulation.

Statistical Thresholds for Concern

Deviations from peer baseline are categorized by statistical distance:

Distance from Peer Mean	Level of Concern	Action
Within 1 SD (normal variation)	None	No action required
1-2 SD above mean	Low	Monitor; document in system
2-3 SD above mean	Moderate	Discuss with manager; may initiate brief investigation
>3 SD above mean	High	Initiate full investigation; consider disciplinary action

Limitations and Cautions

- **Patient Mix Variation:** Even within same unit, patient acuity can vary; accounts for some medication utilization differences
- **Experience Level:** Newer staff may have higher waste rates; should be compared only to similar-experience peers when possible
- **One Month is Not Enough:** A single outlier month does not confirm diversion; 60-90 day patterns are more meaningful
- **Multiple Comparisons:** If many metrics are compared, expect some anomalies by chance; focus on consistent patterns across multiple metrics

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This is a synthetic sample document created for training and demonstration purposes.